#1 Provide the list of markets in which customer "Atliq Exclusive" operates its business in the APAC region.

```
select distinct market

from dim_customer

where customer =  "Atliq Exclusive"

and region = "APAC";
```

#2 What is the percentage of unique product increase in 2021 vs. 2020? The final output contains these fields: unique_products_2020, unique_products_2021, percentage_chg

```
with cte1 as

	(select

		(select count(distinct p.product_code)

		from dim_product p

		join fact_sales_monthly s

		on p.product_code = s.product_code

		where s.fiscal_year = 2020) as unique_products_2020,


		(select count(distinct p.product_code)

		from dim_product p

		join fact_sales_monthly s

		on p.product_code = s.product_code

		where s.fiscal_year = 2021) as unique_products_2021)


select unique_products_2020, unique_products_2021,
```

```sql
round(((unique_products_2021 - unique_products_2020)/unique_products_2020)*100,2)
as percentage_chg

from cte1;
```

#3 Provide a report with all the unique product counts for each segment and sort them in descending order of product counts. The final output contains 2 fields:

segment, product_count

```sql
select segment,

count(distinct product_code) as product_count

from dim_product

group by segment

order by product_count desc;
```

#4 Follow-up: Which segment had the most increase in unique products in 2021 vs 2020? The final output contains these fields:

segment, product_count_2020, product_count_2021, difference

```sql
with products_2020 as

        (select p.segment,

        count(distinct p.product_code) as product_count_2020

        from dim_product p

        join fact_sales_monthly f

        on p.product_code = f.product_code

        where f.fiscal_year = 2020

        group by p.segment),
```

```
products_2021 as
    (select p.segment,
    count(distinct p.product_code) as product_count_2021
    from dim_product p
    join fact_sales_monthly f
    on p.product_code = f.product_code
    where f.fiscal_year = 2021
    group by p.segment)

select
x.segment,
x.product_count_2020,
y.product_count_2021,
(y.product_count_2021 - x.product_count_2020) as difference
from products_2020 x
join products_2021 y
on x.segment = y.segment
order by difference desc;
```

#5 Get the products that have the highest and lowest manufacturing costs. The final output should contain these fields: product_code, product, manufacturing_cost

```
with high as(
select
    p.product_code, p.product,
    m.manufacturing_cost
```

```
from fact_manufacturing_cost m
            join dim_product p
            on m.product_code = p.product_code
where m.manufacturing_cost =
            (select max(m.manufacturing_cost)
            from fact_manufacturing_cost m)),

low as(
select
      p.product_code, p.product,
      m.manufacturing_cost
from fact_manufacturing_cost m
            join dim_product p
            on m.product_code = p.product_code
where m.manufacturing_cost =
            (select min(m.manufacturing_cost)
            from fact_manufacturing_cost m))

select product_code, product, manufacturing_cost
from high h
union
select product_code, product, manufacturing_cost
from low l;
```

#6 Generate a report which contains the top 5 customers who received an average high pre_invoice_discount_pct for the fiscal year 2021 and in the Indian market. The final output contains these fields:

#customer_code, customer ,average_discount_percentage

```
select pre.customer_code, c.customer,
        avg(pre_invoice_discount_pct) as average_discount_percentage
from fact_pre_invoice_deductions pre
        join fact_sales_monthly f
        on f.customer_code = pre.customer_code
        join dim_customer c
  on c.customer_code = pre.customer_code
                where f.fiscal_year = 2021
    and c.market = 'India'
group by pre.customer_code, c.customer
order by average_discount_percentage desc
limit 5;
```

#7 Get the complete report of the Gross sales amount for the customer "Atliq Exclusive" for each month. This analysis helps to get an idea of low and high-performing months and take strategic decisions. The final report contains these columns:

Month, Year, Gross sales Amount

```
select
monthname(f.date) as Month,
f.fiscal_year,
sum(f.sold_quantity*g.gross_price) as Gross_Sales_Amount,
c.customer
from fact_sales_monthly f
```

```
join fact_gross_price g
	on f.fiscal_year = g.fiscal_year
	and f.product_code = g.product_code
join dim_customer c
on f.customer_code = c.customer_code
where c.customer = "Atliq Exclusive"
group by Month, f.fiscal_year, c.customer
order by f.fiscal_year, Month asc;
```

#8 In which quarter of 2020, got the maximum total_sold_quantity? The final output contains these fields sorted by the total_sold_quantity:

Quarter, total_sold_quantity

```
select
case
	when month(date) IN (9,10,11) then "Q1"
  when month(date) IN (12,1,2) then "Q2"
  when month(date) IN (3,4,5) then "Q3"
  when month(date) IN (6,7,8) then "Q4"
end as Quarter,
sum(sold_quantity) as total_sold_qty
from fact_sales_monthly
where fiscal_year = 2020
group by Quarter
order by total_sold_qty;
```

#9 Which channel helped to bring more gross sales in the fiscal year 2021 and the percentage of contribution? The final output contains these fields:

channel, gross_sales_mln, percentage

```
with cte1 as(
select
c.channel,
round(sum(g.gross_price*f.sold_quantity)/1000000,2) as gross_sales_Mln
from dim_customer c
        join fact_sales_monthly f
        on c.customer_code = f.customer_code
  join fact_gross_price g
  on f.product_code = g.product_code
  and f.fiscal_year = g.fiscal_year
where f.fiscal_year = 2021
group by c.channel)

select *,
round(gross_sales_Mln*100/sum(gross_sales_Mln) over (),2) as percentage
from cte1;
```

#10 Get the Top 3 products in each division that have a high total_sold_quantity in the fiscal_year 2021? The final output contains these fields:

division, product_code, product, total_sold_quantity, rank_order

```
with cte1 as(
select
```

```
p.division, p.product_code, p.product,
sum(f.sold_quantity) as total_qty
from dim_product p
join fact_sales_monthly f
        on p.product_code = f.product_code
group by p.product_code, p.product, p.division
order by total_qty desc),

cte2 as(
select *,
dense_rank() over (partition by division order by total_qty) as rank_order
from cte1)

select * from cte2 where rank_order <=3;
```